

Effect of muscle length in a handgrip task on corticomotor excitability of extrinsic and intrinsic hand muscles under resting and submaximal contraction conditions

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Abstract

The neurophysiological mechanisms underlying muscle force control for different wrist postures still need to be better understood. To further elucidate these mechanisms, the present study aimed to investigate the effects of wrist posture on the corticospinal excitability by transcranial magnetic stimulation (TMS) of extrinsic (flexor [FCR] and extensor carpi radialis [ECR]) and intrinsic (flexor pollicis brevis [FPB]) muscles at rest and during a submaximal handgrip strength task. Fourteen subjects (24.06 ± 2.28 years) without neurological or motor disorders were included. We assessed how the wrist posture (neutral: 0° ; flexed: $+45^\circ$; extended: -45°) affects maximal handgrip strength (HGS_{max}) and the motor evoked potentials (MEP) amplitudes during rest and active muscle contractions. HGS_{max} was higher at 0° (133%) than at -45° (93.6%; $p < 0.001$) and $+45^\circ$ (73.9%; $p < 0.001$). MEP amplitudes were higher for the FCR at $+45^\circ$ (83.6%) than at -45° (45.2%; $p = 0.019$) and at $+45^\circ$ (156%; $p < 0.001$) and 0° (146%; $p = 0.014$) than at -45° (106%) at rest and active condition, respectively. Regarding the ECR, the MEP amplitudes were higher at -45° (113%) than at $+45^\circ$ (60.8%; $p < 0.001$) and 0° (72.6%; $p = 0.008$), and at -45° (138%) than $+45^\circ$ (96.7%; $p = 0.007$) also at rest and active

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conditions, respectively. In contrast, the FPB did not reveal any difference among wrist postures and conditions. Although extrinsic and intrinsic hand muscles exhibit overlapping cortical representations and partially share the same innervation, they can be modulated differently depending on the biomechanical constraints.

KEYWORDS

handgrip strength, motor evoked potentials, transcranial magnetic stimulation, wrist joint

1 | INTRODUCTION

Joint postural changes can alter muscle lengths and, consequently, their capacity to generate force. The well-known underlying mechanisms that explain such behavior in muscle performance are based on the crossbridge cycle and the length–tension curve, which determine the interaction level between actin–myosin filaments.¹ In turn, depending on the force needed to regulate one or more joints, the central nervous system (CNS) will adopt different drive strategies to recruit and grade motor unit firing rate to optimize muscle force production.² However, the neurophysiological mechanisms that underlie such muscle gradation force processes on different muscle lengths are not yet fully understood.

Previous studies have reported that postural changes and different muscle lengths can lead to corticospinal modulation of motor evoked potentials (MEP) elicited by single transcranial magnetic stimulation (TMS) pulses.^{3,4} Notably, certain findings provide evidence for remote corticospinal modulation effects on intrinsic muscles, such as the abductor digiti minimi (ADM),⁵ and on extrinsic muscles, such as the flexor and extensor carpi radialis (FCR and ECR),⁶ resulting from shoulder adjustments. In contrast, previous studies have demonstrated a lack of significant remote corticospinal modulation effect on the abductor pollicis brevis (APB), an intrinsic hand muscle, under postural wrist changes.⁷ Divergence in these findings may suggest that the presence or not of a remote corticospinal modulation might be task-dependent,^{8,9} which may help to clarify muscle recruitment patterns and functional synergies in rehabilitation and sports performance.

Regarding manual task performance, the maximum handgrip strength (HGS_{max}), adopted in sports and clinical routines as an evaluation and prognostic index,^{10,11} is influenced by the posture of the wrist joint.¹² On this premise, HGS_{max} is conventionally performed between 0° (neutral position) and 35° of wrist extension^{12–14} as the extrinsic hand muscles, such as the superficial and deep muscle fingers flexors, have their length modified by the angle between the hand and forearm. Unlike a

simple finger action such as thumb flexion, the performance of a complex manual task as a handgrip may lead to corticospinal modulation of intrinsic and extrinsic muscles, even without changes in their length. From this perspective, a handgrip may generate greater corticospinal facilitation of the synergistic muscle network involved in the task than a uni-articular one.⁸ Nevertheless, to the best of our knowledge, no previous study has evaluated whether the corticospinal excitability of extrinsic and intrinsic hand muscles is similarly modulated under different wrist postures during relaxed and contracted conditions while performing a handgrip task. Based on this new knowledge, it would be possible to comprehend better the interplay between the neural control of handgrip and the mechanical properties of muscle tissue under different conditions.

Therefore, the objective of the current study was to investigate the corticospinal excitability of intrinsic and two extrinsic hand muscles under both resting and contracted conditions at different wrist postures/angles (0°, +45°, and –45°). We hypothesized that different wrist postures would result in varying strength levels in HGS_{max} . Additionally, we expected to find a relationship between MEP amplitude and posture, whereby postures that generate muscle shortening would increase MEP amplitude. Further, we posit a corticospinal modulatory effect of the intrinsic hand muscle even if it has not modified its length.

2 | MATERIALS AND METHODS

2.1 | Subjects

Fourteen male subjects (age: 24.06 ± 2.28 years; height: 1.74 ± 0.07 m; body weight: 76.71 ± 8.14 kg) without any neurological or motor disorders were included in the present study. Handedness was assessed with the modified version of the Edinburgh Handedness Inventory (<http://www.brainmapping.org/shared/Edinburgh.php>). Subjects were considered consistently right-handed ($+68.1 \pm 15.3$).

The local ethical committee approved the study (CAAE: 23410513.0.0000.5407). All subjects underwent TMS according to the safety screening questionnaire for TMS and signed informed consent before the experimental session.

2.2 | Instrumentation

2.2.1 | Dynamometry

The handgrip tasks were performed using a handgrip dynamometer (Cefise—Sport Biotechnology; resolution 0.1 kgf or 1 N; weight 1.5 kg; total capacity 100 kgf; dimensions: 230×135×45 mm), and the data were acquired using the N2000 PRO system (Cefise—Sport Biotechnology). To assess the maximum handgrip strength (HGS_{max}) in various wrist postures, a 46 mm fixed handle was utilized. The assessment process involved seating a volunteer comfortably on a chair with their forearms supported on a wooden structure. The right wrist was then positioned at three different angles: flexion (+45°), extension (−45°), and neutral (0°). The wrist angles were identified based on the angle between the radial styloid process and the center of the dynamometer's handle.

The subjects were instructed to maintain the wrist posture while performing maximum handgrip force on the dynamometer. Appendix S1 provides additional details regarding this experimental step. Each trial lasted 5 s, with three attempts at each angle performed in a pseudo-randomized order. The mean of the three tests for each wrist angle was defined as the HGS_{max} .

2.2.2 | Surface EMG recording

Surface electromyographic (sEMG) signals were recorded from the flexor pollicis brevis (FPB), an intrinsic hand muscle, and two extrinsic hand muscles (flexor carpi radialis [FCR] and extensor carpi radialis [ECR]) from the dominant side. MEPs were recorded at rest and during a submaximal handgrip strength task (5% of HGS_{max} : $HGS_{5\%}$) at the three different wrist angles. Surface electrodes (silver/silver chloride [Ag-AgCl]; 1 cm diameter; 2223 BRQ-3M) were placed in a pseudo-monopolar montage with one electrode over the muscle's innervation zone¹⁵ and the other over the nearest bony prominence. The skin was shaved and washed with neutral soap before placing the electrodes. Electrode placements were based on visual identification using isometric contractions and anatomical references suggested by the Atlas of Innervation Zones.¹⁵ Additional details regarding electrode

placement and signal acquisition system characteristics are available in Appendix S1.

To check whether fluctuations in baseline myoelectric activity could limit our interpretation of the results, we measured the mean root mean square (RMS) values in a 500 ms window before the MEPs using a custom Python script (<https://github.com/rmatsuda/RMS-noise-TMS-MEPs>). It helped to control the baseline myoelectric activity and ensure a good signal-to-noise ratio. As a result, we obtained higher RMS values in all three monitored muscles during the active condition. The RMS values (μV ; mean $\pm$ standard deviation) were: FCR: 8.6 ± 4.9 and 25.3 ± 14.0 ; ECR: 29.9 ± 13.3 and 68.9 ± 19.9 ; FPB: 54.6 ± 33.4 and 145.0 ± 57.2 , under resting and active handgrip conditions, respectively.

2.2.3 | Transcranial magnetic stimulation

Corticospinal excitability was assessed using single TMS pulses delivered with a figure-of-eight coil (model FEC-01-100, 75 mm in diameter) connected to a Magstim 200² stimulator (The Magstim Company Ltd., Spring Gardens). Coil positioning was guided using the neuronavigation software InVesalius Navigator (<https://github.com/invesalius/invesalius3>)¹⁶ connected to the motion tracker Fasttrak system (Polhemus). The FCR muscle hotspot was identified as the TMS coil location on the participant's dominant hemisphere (left), eliciting the highest peak-to-peak MEP amplitudes for single-pulse TMS with a fixed supra-threshold intensity. We chose this hotspot because of the important function of the FCR as a wrist flexor¹⁷ and the large overlap of the forearm and hand cortical representations.¹⁸ The hotspot search was performed with the coil tangential to the scalp and approximately perpendicular to the central sulcus (45° to the sagittal plane), inducing a posterior–anterior current on the cortical surface.¹⁸ The hotspot was marked on a cloth cap and digitized with the neuronavigation system to help the experimenter keep the same TMS coil position and orientation during the experimental session. The resting motor threshold (rMT) of the FCR was defined as the lowest intensity of the TMS stimulator that evoked at least three out of six MEP amplitudes equal to or greater than 100 μV .¹⁸ This protocol was implemented to reduce variability and increase the consistency of the obtained MEPs. The subjects were instructed to keep the forearm and wrist in the neutral position, that is, with a 0° angle between the radial styloid process and the center of the dynamometer handle during the hotspot and rMT determinations. MEPs were obtained with stimulation intensity at 120% of the rMT applied to the FCR hotspot during data acquisition.

2.3 | Experimental procedures

In both resting and handgrip contraction conditions, subjects sat with shoulders abducted at 30°, and elbows flexed at 90° in front of a monitor with visual feedback of HGS (for the active state) to maintain $HGS_{5\%}$. A wooden holder kept the dynamometer and forearms positioned with the right wrist at the postures 0°, +45°, and -45° (Figure 1). Further details regarding the anatomical references adopted as parameters for the adjustments of the three wrist joint angles are also available in Appendix S1. In both conditions, the subjects kept their hand holding the dynamometer (resting condition without gripping), equalizing (as much as possible) the tactile feedback between the active and resting conditions.

First, we collected the HGS_{max} with the wrist positioned at each of the three wrist postures. The HGS_{max} allowed us to determine the relative load levels ($HGS_{5\%}$) from each wrist angle, which were required for the active handgrip condition. When compared using a one-way ANOVA, the absolute $HGS_{5\%}$ values used in the different wrist postures were significantly different ($F_{2,25.7}=13.8$; $p<0.001$). Tukey's post-hoc test verified that the mean values across subjects were higher at 0° (18.6 ± 4.29) than at -45° (13.5 ± 4.89 ; $p=0.010$) and +45° (10.5 ± 3.75 ;

$p<0.001$). Then, we recorded between 20 and 30 MEPs, with intervals between single pulses of 5 and 10s in each following condition: three at rest under different wrist angles (0°, +45°, and -45°); and three during a handgrip task at $HGS_{5\%}$ using the same wrist angles. The order of the six conditions was pseudo-randomized for each volunteer. The myoelectric activity in each condition was continuously recorded for 3'30" to collect the MEPs. Subjects were given a break of at least 2 min between conditions to prevent fatigue or reduced alertness. Since the application of the pulses depended on the subject maintaining the target force (active condition), the interval between pulses needed to vary between 5 and 10s, generating between 20 and 30 MEPs per condition.

2.4 | Data and statistical analysis

The sEMG signals were processed using the Signal Hunter¹⁹ software (Figure 2) (<https://github.com/biomaglab/signalhunter>). The MEPs were extracted within a time window between 15 and 60 ms after the TMS pulse to quantify their peak-to-peak amplitudes. The absolute MEP amplitudes for each muscle in each wrist posture and condition (resting and contraction) are presented in

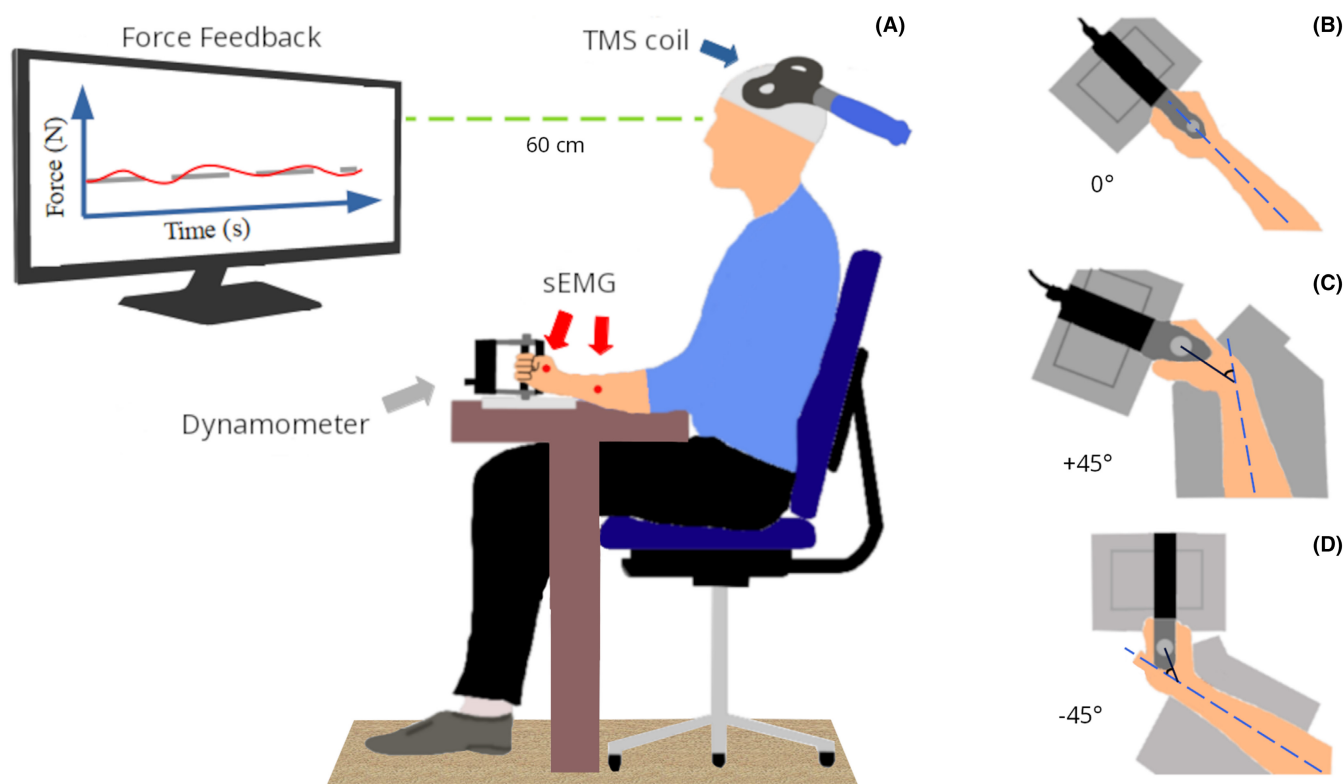


FIGURE 1 The experimental setup was represented schematically as follows: (A) Subjects were seated with their forearms positioned on a wooden holder in a relaxed state. A computer screen was placed in front of them, providing visual force feedback during the active task, which involved exerting 5% of HGS_{max} ($HGS_{5\%}$). The neutral (0°) (B), flexion (+45°) (C), and extension (-45°) (D) postures/angles were applied to the wrist joint during the MEP recording under both rest and active conditions, as observed from a top view.

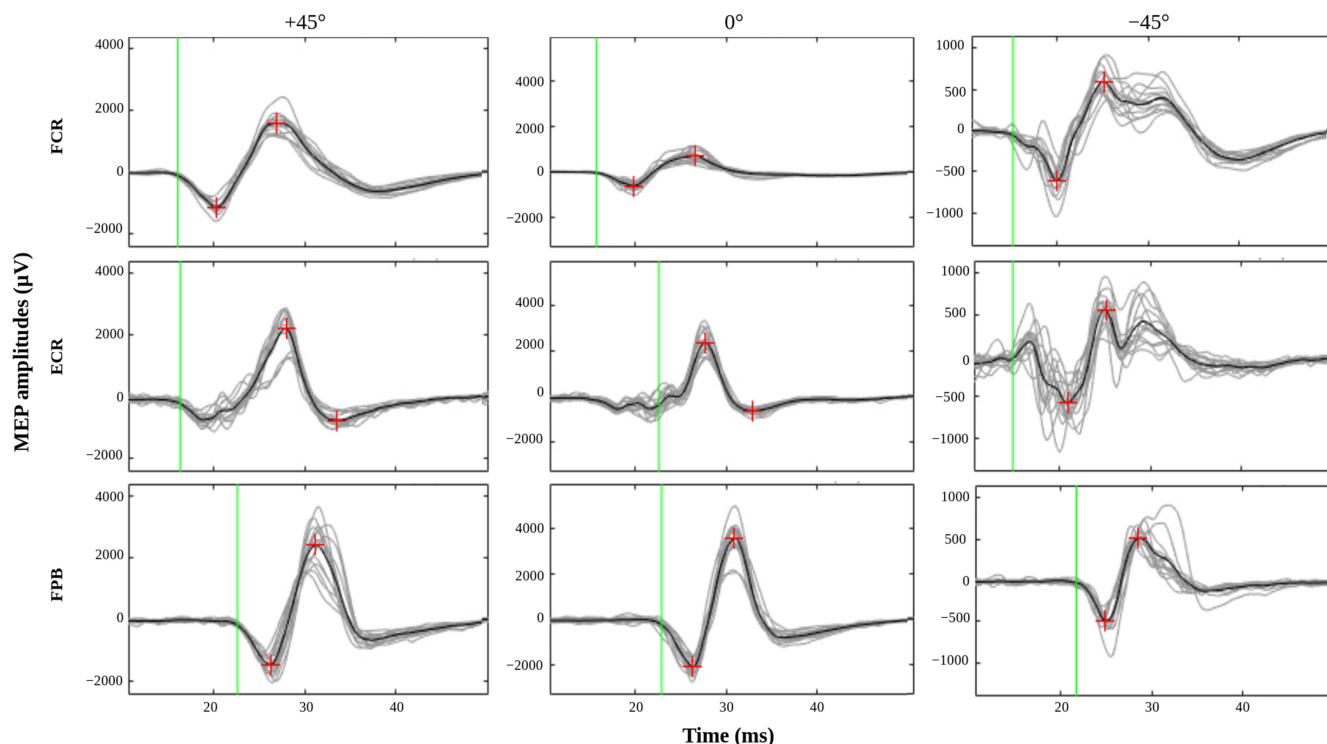


FIGURE 2 An illustrative example of MEP profiles recorded from a volunteer during the HGS_{5%} task. The MEPs are arranged in rows (muscle) and columns (wrist angle). The gray lines represent all MEP trials, and the black line represents the mean MEP profile across the corresponding trials. The peak-to-peak amplitude of the mean MEP is indicated with red crosses and the latency with a green vertical line.

Appendix S2. First, the mean MEP amplitude across trials was calculated for each condition and each subject. Then, the relative values of HGS_{max} and MEP amplitudes were obtained using the ratio between every variable obtained in the test (HGS_{max} or mean MEP amplitude) and the mean across all conditions of the same variable (e.g., (mean MEP amplitude/mean MEP amplitudes of all conditions) × 100). Finally, these relative average values were used for statistical analysis.

The statistical analysis was done in the software Jamovi (version 2.3). To assess whether the data met the assumptions for independent measures analysis of variance (ANOVA), data distribution were visually inspected using Q–Q plots and Levene's test was applied to verify the homogeneity of variances. We used a one-way ANOVA to assess the effect of wrist angle (+45° [flexion], −45° [extension], and 0° [neutral]) on the HGS_{max}. We used a two-way ANOVA to assess the effect of wrist angle and task condition (rest and HGS_{5%}) on the MEP amplitude. The comparisons of HGS_{max} and MEP amplitude between the rest and handgrip conditions were determined using the mean across the three angles at rest multiplied by the mean across the three angles during handgrip. Post-hoc multiple comparisons were performed with Tukey's post-hoc test with correction for family-wise error rate. Our results are presented as mean (±standard deviation). The *p* value was set at 0.05.

3 | RESULTS

3.1 | Maximal handgrip strength (HGS_{max})

There was an effect of the wrist angle on the HGS_{max}. The relative HGS_{max} was significantly greater at 0° (133 ± 18.6%) than −45° (93.6 ± 18.4%; *p* < 0.001) and +45° (73.9 ± 16.1%; *p* < 0.001) ($F_{2,25,9} = 39.0$; *p* < 0.001). Additionally, −45° was significantly greater than +45° (*p* = 0.015; Figure 3).

3.2 | MEP amplitude

A two-way ANOVA was performed to compare the effect of wrist angle, condition, and the interaction of both factors on the relative MEP amplitude for each muscle (Figure 4). The relative MEP amplitude varied depending on the wrist angle ($F_{2,78} = 14.21$; *p* < 0.001) for the FCR muscle. At rest, Tukey's test for multiple comparisons revealed that the mean MEP amplitude was significantly greater at +45° (83.6 ± 23.3%) than at −45° (45.2 ± 16.7%; *p* = 0.019). During the active condition, the MEP amplitude was significantly greater at +45° (156 ± 30.3%; *p* < 0.001) and 0° (146 ± 39.1%; *p* = 0.014) than at −45° (106 ± 44.7%). Also, the MEP amplitude in FCR was greater in the active (136 ± 43.4%) than in the resting condition (61 ± 26.7%;

$F_{1,78}=122.52$; $p<0.001$). There was no significant interaction between wrist angle and condition ($F_{2,78}=1.73$; $p=0.183$).

In the ECR muscle, the relative MEP amplitude also varied depending on the wrist angle ($F_{2,78}=17.11$; $p<0.001$). At rest, the mean MEP amplitude was significantly greater at -45° ($113\pm27.5\%$) than at $+45^\circ$ ($60.8\pm19.2\%$; $p<0.001$) and 0° ($72.6\pm29.0\%$; $p=0.008$). During the active handgrip condition, we found that the MEP amplitude at -45° ($138\pm40.7\%$) was significantly greater than $+45^\circ$ ($96.7\pm28.2\%$; $p=0.007$). In addition, there was an effect of condition ($F_{1,78}=30.05$; $p<0.001$), where the MEP amplitude in active ($119\pm37.9\%$) was greater than at rest

($82.3\pm33.8\%$). Again, there was no significant interaction between wrist angle and condition ($F_{2,78}=1.04$; $p=0.358$).

Finally, the relative MEP amplitudes of the FPB were not different in the different wrist postures/angles ($F_{2,78}=1.66$; $p=0.198$). However, the MEP amplitude was significantly greater during active ($128\pm41.5\%$) than at rest ($71.1\pm43.7\%$; $F_{1,78}=40.54$; $p<0.001$). In addition, there was a significant interaction between wrist angle and condition ($F_{2,78}=3.68$; $p=0.030$), showing that the effects of wrist posture depended on the handgrip condition. We performed a post-hoc analysis to evaluate the differences in MEP amplitudes from the FPB muscle across the tested conditions. We observed significant higher MEP amplitudes at 0° -active (131 ± 48.3 ; $p<0.001$), -45° -active (131 ± 44.9 ; $p<0.001$), and $+45^\circ$ -active (122 ± 31.9 ; $p=0.002$) compared to 0° -rest ($60.1\pm41.6\%$). Moreover, 0° -active MEP amplitudes were significantly higher than -45° -rest (53.7 ± 28.9 ; $p<0.001$). Additionally, -45° -active ($p<0.001$), $+45^\circ$ -rest (99.5 ± 46.0 ; $p=0.046$), and $+45^\circ$ -active ($p<0.001$) MEP amplitudes were significantly higher than at -45° -rest.

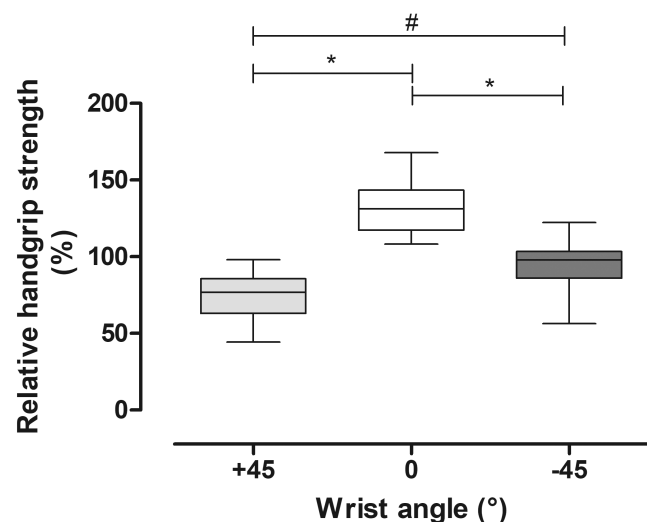


FIGURE 3 Boxplots with minimum, maximum, first, second, and third quartile values of HGS_{max} in the three different wrist angles: extension (-45°), neutral (0°), and flexion ($+45^\circ$) in healthy subjects ($n=14$). The relative values of the HGS_{max} were significantly higher at 0° than at $+45^\circ$ and -45° . (#) $p=0.015$; (*) $p<0.001$.

4 | DISCUSSION

4.1 | Handgrip strength

This study aimed to investigate the effects of wrist angle on handgrip strength capacity. The performance on the HGS_{max} was different for each of the tested wrist joint angles ($+45^\circ$, 0° , and -45°). The neutral posture (0°) led to the maximum handgrip strength, while the wrist flexion at $+45^\circ$ led to the weakest. Previous studies have shown that the wrist angle influences the HGS_{max}^{12,13,20}. The optimal wrist posture for achieving HGS_{max} is characterized by wrist extension, typically within a range of 35° , with

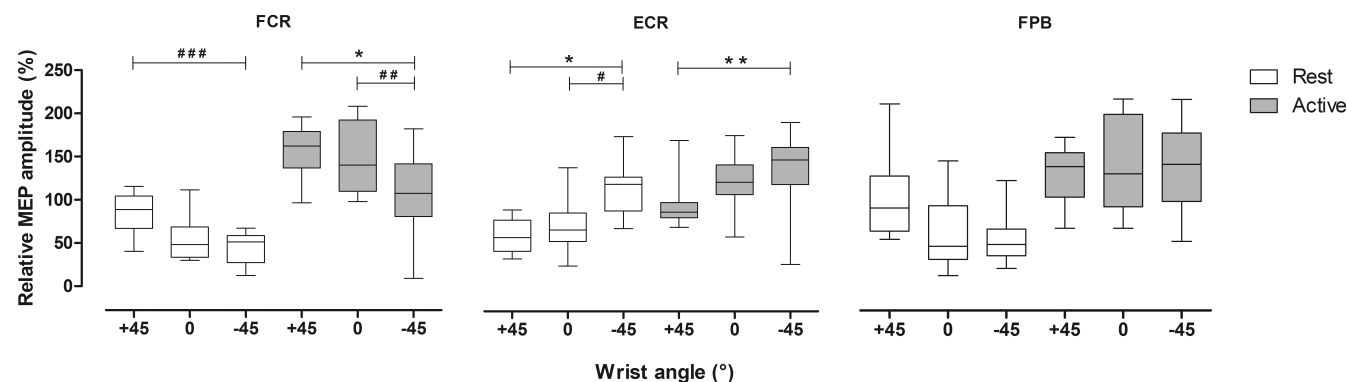


FIGURE 4 Boxplots with minimum, maximum, first, second, and third quartile values of relative MEP amplitude values for FCR, ECR, and FPB muscles at rest and during the active handgrip conditions (HGS_{5%}) in the three different wrist angles: extension (-45°), neutral (0°), and flexion ($+45^\circ$) in healthy subjects ($n=14$). Statistically significant differences are represented as (*) $p<0.001$, (**) $p=0.007$, (#) $p=0.008$, (##) $p=0.014$ and (###) $p=0.019$.

minor variations due to ulnar deviation.^{12,13} It supports our findings and indicates that wrist flexion and extension greater than 40° lead to lower performance in maximal handgrip strength. It is worth noting that methodological factors, such as the control of radial-ulnar deviations and handle length, can pose challenges in comparing results, as O'Driscoll shows.¹² To examine the contribution of the FPB muscle in opposing the four fingers during handgrip, we utilized the smallest feasible grip with our equipment in this investigation. Utilizing a fixed handle throughout the three postures and relying on relative HGS values potentially reduced these issues.

It is well-known that angular changes in one or more joints can alter the muscle capacity to generate force. This premise is based on the length versus tension relationship curves.¹ The reduction in HGS_{max} observed at +45° and −45° is attributed to decreased mechanical efficiency during the task, as the rate of crossbridge formation is reduced due to the change in length of the finger's superficial and deep flexor muscles.²¹

4.2 | MEP amplitude modulation at rest

We observed that the MEP amplitude at rest was greater at wrist angles that shortened muscle length: flexion for the FCR and extension for the ECR; and smaller when the muscle length was increased: extension for the FCR and flexion for the ECR. Lewis et al.⁷ observed that wrist flexion led to an increase in MEP amplitude during static and passive postural changes, while wrist extension led to a decrease in MEP amplitude in the FCR muscle. Other authors also reported this effect during passive wrist movement.^{3,22} Forman et al.⁴ suggested that the reduced afferent input of the muscle spindle caused by the change in muscle length would be responsible for modulating the corticospinal excitability. Accordingly, a shortened muscle would evoke a reduction in muscle spindle firing rate, signaling an unfavorable condition for force production and ultimately increasing corticospinal motor output. Therefore, our findings at rest are likely explained by two complementary mechanisms: First, input from visual²³ and proprioceptive receptors inform the CNS about the muscle status to generate force and contribute to the movement.^{4,24} Direct evidence of the role of the muscle spindle in monitoring muscle length to update the motor cortex and spinal motoneurons about joint position sense has already been brought forward. This evidence includes maintenance of position sense and motion even after ligament and joint capsule replacement, dependence on the integrity of afferent projections from muscle spindles for joint position sense, and errors in position sense caused by muscle vibrations.²⁴ Second, by reaching the motor cortex,

these sensory inputs could increase the corticospinal activity, making a TMS pulse capable of activating a broader population of spinal²⁵ and cortical neurons.^{4,26} Thus, we could detect an increase in MEP amplitude due to increased excitability of the corticomotor pathway.

We also investigated whether corticospinal excitability modulation would occur in the intrinsic hand muscle (FPB, supplied by the median nerve), even without changing its length on different wrist angles. The FPB did not show similar modulation to that observed for the FCR, that is, larger MEP amplitudes at wrist flexion and smaller MEP amplitudes at wrist extension. It indicates that the modulation of the corticospinal excitability of the FPB, which has not changed its length, is not entirely related to changes in the excitability of the FCR. Mogk et al.²⁷ found corticospinal modulation even without changes in muscle length (e.g., the modulation of the corticospinal excitability of the posterior deltoid muscle derived from a change in forearm posture). These findings are also corroborated by Forman et al.,⁶ who showed modulation in the corticospinal excitability of the forearm muscles in different shoulder postures.

On the other hand, Ginanneschi et al.²⁸ have observed corticospinal modulation in an intrinsic hand muscle after changing the shoulder posture. These findings suggest that proprioceptive receptors from more proximal joints, such as the shoulder, can modulate the cortical and spinal excitability of intrinsic hand muscles in addition to muscle length. Thus, it was expected that changes in wrist posture could modulate the corticospinal excitability of the FPB, mainly because FPB and FCR share the same innervation. Additional hypotheses regarding the shared innervation of FPB and FCR muscles and its potential contribution to their corticospinal modulation²⁹⁻³¹ can be found in the Appendix S3.

4.3 | MEP amplitude modulation during handgrip task

In the present study, MEP amplitudes of the FCR and ECR muscles were similarly modulated in each wrist angle during both conditions (rest and HGS_{5%}). As well as at rest, this similarity indicates that corticospinal excitability also adapts to different muscle lengths during active situations of low force production (5% of maximum voluntary isometric contractions [MVIC]). However, studies along these lines have shown controversial results. For example, Collins and Button³² observed smaller MEP amplitudes in the *biceps brachii* with the shoulder at 90° of flexion (shortened muscle) than 0° during a contraction (10% of MVIC). Some studies have observed no effect of muscle length on MEP amplitudes in voluntary conditions.^{4,33} Conversely, others corroborate our findings and show greater MEP amplitudes in muscle-shortening postures during voluntary contraction.^{3,34}

The mechanisms presented to explain the modulations observed during rest, such as greater mobilization of cortical neurons and reduced firing rate of the muscle spindle during muscle shortening, could also explain the results during the handgrip task. These mechanisms, in theory, can be even more effective as the kinesthetic input caused by passive movements (and possibly also during active conditions) is higher than in the resting state, as already reported.²³ As for the conflicting results of corticospinal excitability in different muscle lengths during active conditions, they may be the result of other experimental conditions, such as the type of muscle length manipulation (active $\times$ passive $\times$ static, differences in the rate of change of length as a function of time), differences in active conditions (precision grip, handgrip, uni-articular movements), differences in EMG and TMS settings, investigated muscle, or even the variability of the corticospinal excitability and MEP amplitude itself.

The present study did not observe corticospinal excitability modulation in FPB under active conditions across wrist postures. Forman et al.⁶ observed that neuron reactivity for the FCR at 5% of maximal voluntary excitation (peak myoelectric activity of the maximal voluntary handgrip force) was significantly higher with the shoulder abducted than flexed. Furthermore, the corticospinal excitability of the ECR was markedly higher at 45° than at 90° and 120° shoulder flexion at 30% of the maximal voluntary handgrip. Interestingly, Gagné and Schneider⁹ investigated the effects of wrist angle changes on the corticospinal excitability of the first dorsal interosseous (FDI) muscle during a precision grip task. They found that modulations in FDI were only observed during active and passive wrist flexion and extension but not under static conditions. These findings, although somewhat controversial, suggest that changes in the length of forearm muscles that share innervation with the FDI, such as the ulnar flexor carpi, may have some effect on proximal-distal synergistic modulation. Perez and Rothwell³⁵ also observed corticospinal excitability modulation in the FDI in different postures (neutral, pronated, and supinated forearm) during the precision grip of a small cylinder but not in the other two intrinsic hand muscles (ADM and APB).

These and other studies suggest corticospinal excitability modulation under volitional conditions could be task and muscle dependent.^{34,36} Hasegawa et al.³⁶ have indicated that precision and power grip tasks may recruit different corticospinal circuits, where precision grip tasks would induce greater MEP amplitude amplitudes in intrinsic hand muscles. Therefore, we hypothesize that during a fine grasp task, where there is greater activation of cortical and spinal neuron populations, there could be greater synergistic recruitment between FCR and FPB. Precision tasks elicit greater activity of monosynaptic corticomotor neurons, leading to a larger neuronal population being activated in M1 through TMS. Additionally, there is a greater size and degree of

connectivity between monosynaptic corticomotor neurons in such tasks.^{36,37} It is worth mentioning that FPB participation has been previously observed during hand grip tasks involving objects such as a cylinder.³⁸ However, in our handgrip task, the volunteer was not required to stabilize the dynamometer, which may have contributed to the reduced FPB contribution. Nevertheless, if this was a determining factor, we believe that modulations in the excitability of the FPB muscle would have been detected, at least while at rest.

Using a handgrip task, no synergistic modulation was observed between FCR and FPB during changes in wrist angle. It is also possible that HGS_{5%} was too low to observe these effects. Further studies are needed, possibly using a precision task, monitoring other intrinsic hand muscles and/or using a higher intensity task.

This study showed for the first time that during a low-intensity handgrip task under different wrist postures, no synergistic facilitation/inhibition was observed between the FCR and FPB. In addition, it brings new findings about the corticospinal excitability of an intrinsic hand muscle that does not modify its length during a handgrip task. This study corroborates previous findings on the influence of muscle length on corticospinal excitability, both at rest and under active conditions. Moreover, these results point to a modulation of the corticospinal pathway whose effects are related to the need to supply eventual mechanical deficits resulting from the muscular length and, therefore, to set up a motor unit recruitment pattern to ensure its effectiveness on generating force.

4.4 | Limitations of the study

Using a standard grip size on the dynamometer may have caused the handle to be positioned over the thenar region for some subjects, leading to decreased FPB function during the handgrip. Additionally, the magnitude of the submaximal contraction was kept low to reduce the influence of oscillations in myoelectric activity on the collected MEPs. Nevertheless, a more intense task could have led to synchronous modulations between the FCR and the FPB.

Furthermore, while it is clear that different wrist postures affect the length of the extrinsic muscles that cross the wrist joint, the magnitude of this change cannot be determined. Additionally, although the FPB does not change its length during a handgrip task, it may have undergone minimal changes in length due to the two degrees of freedom of the trapeziometacarpal joint.³⁹ Finally, the exact position of the thumb was not monitored, making it challenging to ensure that its length remained constant during the experiment.

Despite these limitations, the study significantly contributes to investigating synergistic modulations between proximal and distal muscles during a functional handgrip

task. In summary, the complexity of controlling mechanical variables makes it challenging to compare with uniaxial tasks.

4.5 | Perspectives

The handgrip is a functional task of fundamental importance in daily and sports activities.^{10,11} Sports that necessitate handgrip while adopting various wrist postures (e.g., golf, tennis, table tennis) should consider the presence of inefficient wrist postures that could impact force generation capability. For instance, when maintained for a prolonged time, the increased corticospinal activity of wrist muscles in less efficient postures (e.g., increased FCR sEMG activity in wrist flexion) can result in reduced performance and increased risk of injury. On the other hand, sports that require manual dexterity while maintaining different wrist postures (e.g., archery and sport shooting) may benefit from these findings as different wrist postures do not modify the activation of the intrinsic muscles of the hand. From a clinical point of view, these results may be relevant for patients in rehabilitation after wrist/hand injury or surgery. The therapeutic approach to regain handgrip capacity can initiate with postures that facilitate the corticospinal pathway, thereby promoting muscle recruitment and progressing to more complex postures with higher demands on corticospinal activity (wrist flexion). In addition, given the independence between the function of the flexor muscles of the wrist and the thenar muscles of the hand, rehabilitation programs should encompass the functional recovery of the wrist and thumb in parallel.

In conclusion, this study also reinforces the need for a better understanding of the synergy between the intrinsic and extrinsic muscles of the hand during different postures/angles and wrist conditions in healthy individuals, which may help better contribute to sports performance and rehabilitation programs.

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CONFLICT OF INTEREST STATEMENT

VHM, CDV, BR, VHS, RHM, LAI, and MACG declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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